Joint phasing of sequence variants and DNA methylation from long reads with LongPhase 2

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Abstract

Long-read sequencing captures SNVs, indels, structural variants (SVs) and 5-methylcytosine (5mC) on the same molecules, yet phasers use these variant classes separately. Here we present LongPhase 2, which phases all four in one graph and uses a graph neural network to withhold likely misphased variants. Scored against the telomere-to-telomere (T2T) HG002 assembly on nanopore and PacBio HiFi data at 10–60× coverage, LongPhase 2 had 1.7- to 5.0-fold lower SNV switch error rates than widely used read-based phasers and 1.5- to 5.4-fold lower rates than methylation-aware phasers at every coverage, and phased SVs with 1.3- to 2.3-fold lower error rates than WhatsHap. Co-phasing all four classes lengthened phase blocks by 18–46% while switch error rates stayed at least 2.5-fold below those of WhatsHap. Switch errors shared by all phasers under the earlier GIAB benchmark largely disappeared under the T2T-derived benchmark, which distinguished the phasers more clearly, particularly in segmental duplications and other difficult regions. LongPhase 2 makes joint phasing of genetic and epigenetic variation accurate and fast enough for routine long-read analysis.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Determining whether two variants lie in *cis* or in *trans* is central to identifying compound heterozygosity in recessive disease[1]. Allele-specific methylation on a phased haplotype reveals its parent of origin[2]. Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow[3,4]. Read-based phasing does not require parental sequencing, a population reference panel of the kind statistical phasing depends on[5], or the additional Hi-C or Strand-seq libraries used to phase whole chromosomes[6].

Read-based phasers assign heterozygous alleles to haplotypes by maximizing agreement with the reads, and they differ in both their models and the variant types they phase. WhatsHap solves a weighted minimum-error-correction problem[7] and, by re-aligning each read to both

alleles, phases indels, including large ones, together with single-nucleotide variants (SNVs)[8]. HapCUT2 uses a likelihood-based model and also phases indels[9], Margin phases SNVs with a hidden Markov model[10], and HiPhase jointly phases small variants, structural variants (SVs) and tandem repeats from PacBio HiFi reads[11]. We previously introduced LongPhase[12], which recasts phasing as a weighted allele graph, resolves haplotypes at chromosome scale and co-phases SNVs with SVs[13,14]. Each phaser therefore handles a different subset of variant types, and none phases sequence variants and allele-specific methylation in one model. Read-based phasing is no longer an end point but an intermediate step whose output other tools use directly. Clair3, ClairS and ClairS-TO phase reads with WhatsHap or with LongPhase as part of variant calling[15–18]; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase[19]; and Sniffles2 and Severus call SVs from the haplotagged alignments that these phasers write[13,20]. A phase error therefore does not stay inside the phaser: a single switch reverses the haplotype labels of every read beyond it in the block, and the small-variant, somatic and SV callers that rely on those labels carry the error into their own output.

Methylation offers a further source of phase evidence, but methylation-aware phasers gain contiguity at a possible cost in accuracy. Haplotype-specific CpG methylation (5-methylcytosine, 5mC), which long reads report together with the sequence, can link reads across regions that lack heterozygous sequence variants[21,22]. MethPhaser, HapBridge and LongHap use this signal to join blocks after, or alongside, sequence-based phasing[22–24]; MethPhaser, for example, lengthened phase-block N50 by 78–151% at a reported phasing accuracy of 83.4–98.7%[22]. Because the informative CpGs are learned from reads that the sequence phase has already assigned, its errors propagate into the methylation evidence. Methylation is also a less consistent marker than a sequence allele: haplotype-specific methylation is confined to a small fraction of CpGs, and the state of a CpG can differ between molecules of the same haplotype, varies between cells and tissues, and is called as a per-molecule probability rather than a base[21,22]. A block join that rests on a few such CpGs can reverse the phase of every variant beyond it; methylation therefore needs to be weighed against the sequence alleles on the same reads, not added after the phase is fixed.

Phase errors are not spread evenly across the genome. Base-calling errors cluster in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages between heterozygous sites, and heterozygous calls in copy-number-altered regions can mislead the phasing graph[25,26]. Current phasers weight read observations by base quality[7,9], but a high-quality base on a misplaced read still supports a false linkage. Phasers also give little indication of which phase assignments are wrong. WhatsHap and Margin report no per-variant confidence, and LongPhase reports only read support. HapCUT2 prunes variants with low posterior confidence[9], but misplaced reads that agree with each other can keep this confidence high. Learning-based phasers learn to optimize the phasing objective, not to detect its errors[27,28]. A phaser therefore needs both to weight evidence by its sequence context and to mark the phase assignments that remain unreliable.

Phasers are commonly benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38[29,30]. The HG002 v4.2.1 release excludes both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications in which

mis-mapped reads and collapsed paralogues create false linkages[25,26,30,31]. It also phases only the 74% of heterozygous variants on which trio-based and Strand-seq phasing agreed[30]. Expanding GIAB from 85% to 92% of GRCh38 revealed eightfold more false negatives in a short-read call set[30]. The recent complete diploid assembly of HG002, T2T-HG002, phases every heterozygous site on both haplotypes of every chromosome, including 701 Mb of autosomal sequence absent from v4.2.1[31]. It therefore allows phasing to be assessed in the difficult regions that mapping-based benchmarks exclude.

Here we present LongPhase 2, a read-based phaser that addresses these limitations. It phases SNVs, small indels, SVs and allele-specific 5mC in one weighted graph, so that methylation is weighed against the sequence alleles during haplotype inference. It down-weights read evidence in difficult sequence contexts, and a graph neural network withholds phase assignments that are likely to be wrong. We evaluate it on nanopore and PacBio HiFi data against both the mapping-based and the assembly-based HG002 benchmarks.

Results

LongPhase 2 combines joint phasing with learned error detection

LongPhase 2 takes long-read alignments, a reference genome and heterozygous variant calls, and phases four variant classes together (Fig. 1a). SNVs, small indels and SVs come from upstream variant callers (e.g., Clair3[15], DeepVariant[32] and Sniffles2[13]). LongPhase 2 calls allele-specific 5mC sites from the same reads, keeping CpGs whose methylated and unmethylated reads are balanced and whose methylation co-segregates with a neighbouring heterozygous marker (Methods). Each heterozygous variant and 5mC site contributes a reference and an alternate allele node to one graph per chromosome, and each read links the alleles it carries (Fig. 1b). Methylation is therefore phased together with the sequence alleles of the same reads, instead of being used afterwards to join blocks whose sequence phase is already fixed. SVs inform phasing within blocks but never connect two blocks on their own.

Read evidence is weighted by its quality and its sequence context (Fig. 1b). Observations at low-quality bases and indels in tandem repeats are down-weighted, because both can link the wrong alleles. A copy-number-aware filter removes variants whose alternate-supporting reads carry excess heterozygous mismatches in intervals bounded by clipped alignments, which marks candidate copy-number-altered regions. Haplotypes are then resolved by multi-neighbour voting: each variant votes for an allele pairing with each downstream neighbour, and a pair without a clear majority casts no vote. A second, read-based pass assigns reads to haplotypes and leaves a variant unphased when the two read populations disagree on its phase.

LongPhase 2 then scores the uncertainty of every phase assignment and re-examines the least certain ones (Fig. 1c). Uncertainty is measured as phasing entropy, from 0 bits when all haplotype support agrees to 1 bit for an even split. Around each high-entropy variant, a graph neural network (GNN) reads the local phasing graph, combining message passing along its edges with self-attention across the window, and predicts where the phase starts to disagree with that of the neighbouring variants. Predicted variants are unphased rather than flipped (Fig. 1d), so an incorrect prediction cannot add a switch error.

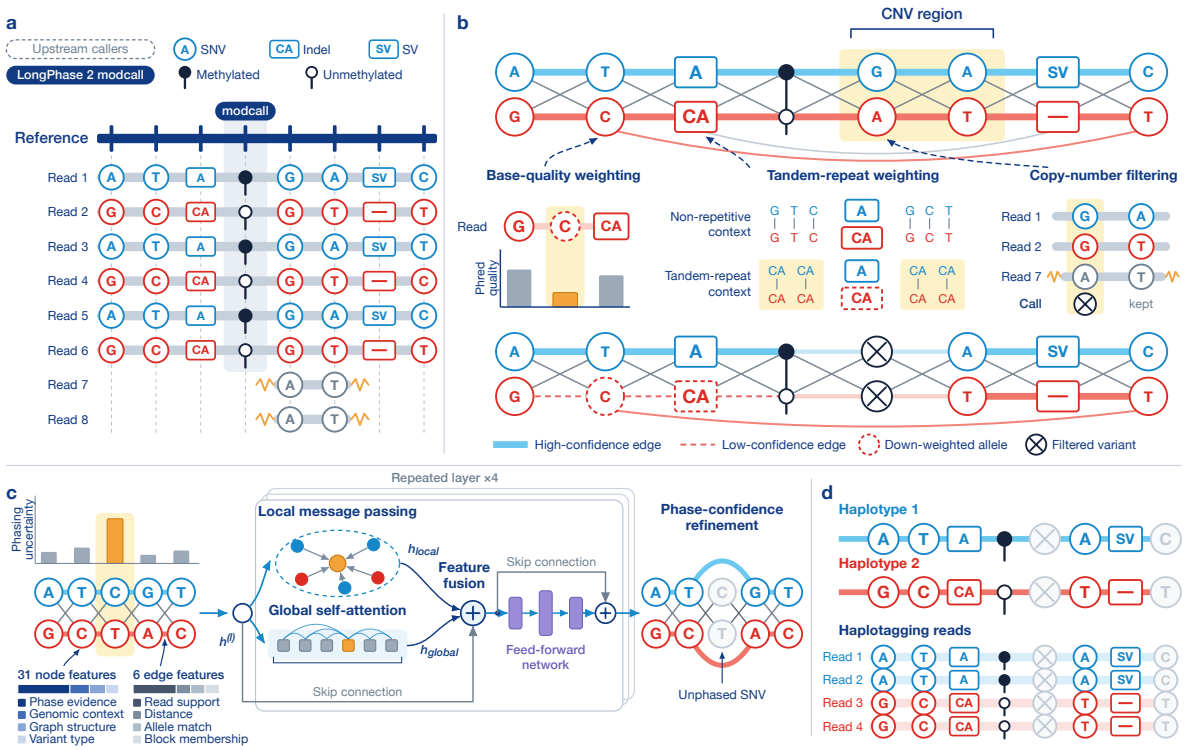


Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels and SVs from upstream variant callers (dashed tag) and 5mC sites called by the `modcall` module of LongPhase 2 (navy tag), carried by the same long reads, enter one phasing problem; alleles are coloured by read haplotype, a dash marks the reference allele of the SV, and grey reads are clipped at both ends (orange). **b**, Weighted phasing graph before (top) and after (bottom) evidence weighting. Read-supported edges link allele nodes; observations at low-quality bases and indels in tandem repeats are down-weighted, and a variant whose alternate allele comes from clipped reads in a copy-number-altered region is filtered (insets). **c**, A graph neural network evaluates local phasing graphs centred on variants with high phasing uncertainty (Methods). Predicted phase-disagreement onsets are selectively unphased (GNN correction; grey). **d**, Output: phased haplotypes of all four variant classes, with the filtered variant (cross) and the selected variant left unphased (faded), and haplotagged reads.

LongPhase 2 reduces switch error rates across variant classes and sequencing platforms

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 reads at 10–60× and PacBio HiFi reads at 10–50×, with the same alignments and variant calls for every phaser (Methods). Phase was scored against two HG002 truth sets: GIAB v4.2.1, integrated from mapping-based call sets, and GIAB v5.0q, phased by the diploid telomere-to-telomere (T2T) assembly. Unless stated otherwise, results refer to SNV phasing against v5.0q.

In SNV-only phasing of nanopore data, LongPhase 2 consistently achieved lower switch error rates (0.022–0.081%) than WhatsHap (0.081–0.233%) and HapCUT2 (0.104–0.266%) across the tested coverages (Fig. 2a), corresponding to 2.9–5.0-fold reductions (Supplementary Table 1). LongPhase 2 also had 1.3–1.8-fold lower block-wise Hamming distances, the fraction of misassigned variants within blocks, than WhatsHap and HapCUT2 (Supplementary Fig. 1a). When indels were included in the nanopore runs, LongPhase 2 had 3.0–3.7-fold lower SNV switch error rates than WhatsHap and 3.5–4.8-fold lower rates than HapCUT2 (Fig. 2b). The difference was larger for indel pairs, adjacent variants of which at least one is an indel: LongPhase 2 had 8–32-fold lower indel-pair switch error rates than WhatsHap (0.029–0.100% versus 0.75–0.91%; Fig. 2b and Supplementary Table 2). The fold difference increased with coverage because the rate of LongPhase 2 decreased and that of WhatsHap did not. Indel Hamming distances showed the same pattern: LongPhase 2 had 1.8–2.7-fold lower values than WhatsHap (3.5–7.1% versus 9.0–13.8%), and the fold difference increased with coverage (Supplementary Fig. 1b).

On PacBio HiFi data, LongPhase 2 had lower SNV switch error rates (0.033–0.049%) than HiPhase[11] (0.057–0.085%), the phaser of the PacBio whole-genome workflow, WhatsHap (0.077–0.128%) and HapCUT2 (0.096–0.152%) at every coverage (Fig. 2c and Supplementary Table 3), corresponding to 1.7–1.9-, 2.4–2.8- and 2.9–3.5-fold reductions, respectively. The difference from WhatsHap narrowed from 2.6–2.8-fold at 10–30× to 2.4-fold at 40–50×, whereas the difference from HiPhase remained at 1.7–1.9-fold at every coverage. Hamming distances were 0.7–1.2% for all four tools; LongPhase 2 had the lowest values at 10, 30 and 40×, HiPhase at 20× and WhatsHap at 50× (Supplementary Fig. 1c). With indels, LongPhase 2 had lower SNV switch error rates (0.041–0.094%) than HiPhase (0.070–0.164%), WhatsHap (0.096–0.252%) and HapCUT2 (0.131–0.272%) at every coverage (Fig. 2d and Supplementary Table 4), corresponding to 1.6–1.9-, 2.3–2.9- and 2.9–3.5-fold reductions, respectively. Its Hamming distances, which include indels in these runs, were 1.3–2.4-fold lower than those of the other three tools at every coverage (Supplementary Fig. 1d).

Block contiguity was similar among phasers without indels, whereas with indels the other phasers produced longer blocks at higher switch error rates (Fig. 2e–h). In SNV-only phasing of nanopore data, block N50 increased from 0.84–0.95 Mb at 10× to 2.72–2.87 Mb at 60× for all three tools; the blocks of LongPhase 2 were up to 11% shorter than those of WhatsHap and HapCUT2 at 10–20× and 2–5% longer at 30–60× (Fig. 2e and Supplementary Table 1). With indels, the blocks of WhatsHap and HapCUT2 were 1.1–1.3- and 1.3–1.4-fold longer than those of LongPhase 2, at 3.0–3.7- and 3.5–4.8-fold higher SNV switch error rates (Fig. 2f). On HiFi data, block N50 increased from 340 to 580 kb and differed by at most 3% among the four tools (Fig. 2g). With indels, the HiFi blocks of WhatsHap, HapCUT2 and HiPhase were 1–8%,

10–17% and 11–17% longer than those of LongPhase 2 (Fig. 2h), at 2.3–2.9-, 2.9–3.5- and 1.6–1.9-fold higher switch error rates. HiFi blocks were 2.5–4.3-fold shorter than nanopore blocks, consistent with the shorter reads.

Joint phasing extends lower error rates to SVs and 5mC

We next asked whether the lower error rates of LongPhase 2 extend to SVs. Because SVs are sparse, we scored each heterozygous deletion and insertion of at least 50 bp called by Sniffles2 separately, not as adjacent variant pairs. An SV was wrongly phased when it was placed on a different haplotype from that in the T2T-HG002 assembly (Methods). On nanopore data, LongPhase 2 had 1.6–2.1-fold lower SV phasing error rates than WhatsHap at every coverage (5.8–11.3% versus 11.2–18.7% at 10–60×; Fig. 3a and Supplementary Table 5). We found that WhatsHap made two types of SV error more often than LongPhase 2. It placed 1.8–2.3 times as many heterozygous SVs on the wrong haplotype and phased 2.7–3.3 times as many SVs that are homozygous in the assembly, which are genotype errors of the variant caller. On HiFi data, LongPhase 2 had 1.4–1.5-fold lower SV phasing error rates than WhatsHap at every coverage (5.8–6.6% versus 8.9–9.4%; Fig. 3b). HiPhase had SV phasing error rates similar to those of LongPhase 2 (5.4–6.4% versus 5.8–6.6%). With indels co-phased, LongPhase 2 had 1.7–2.3-fold lower SV phasing error rates than WhatsHap on nanopore data and 1.3–1.7-fold lower rates on HiFi data, and rates similar to those of HiPhase (Supplementary Fig. 2). For SVs of at least 1 kb, LongPhase 2 had the lowest SV phasing error rates at every coverage on both platforms (Fig. 3c): 8.0–12.8% on nanopore data, compared with 15.2–20.7% for WhatsHap, and 7.8–9.1% on HiFi data, compared with 10.9–13.3% for HiPhase and 13.1–14.3% for WhatsHap. For SVs of 50 bp to 1 kb on HiFi data, HiPhase had the lowest SV phasing error rates (4.8–5.8%, compared with 5.6–6.5% for LongPhase 2; Supplementary Table 5).

We then asked whether phasing 5mC together with sequence variants is more accurate than adding methylation after phasing. MethPhaser[22] adds methylation after phasing: given the SNV phasing from which the joint SNV and 5mC phasing of LongPhase 2 started (Methods), it joins the blocks of that phasing without changing phase within them. On nanopore data, LongPhase 2 had the lowest switch error rate of the three phasers that use 5mC at every coverage, 1.5–1.8-fold lower than that of MethPhaser and 2.2–5.4-fold lower than that of LongHap[24] (Fig. 3d and Supplementary Table 4). MethPhaser and LongHap produced longer blocks than LongPhase 2 at most coverages (Fig. 3e and Supplementary Table 4). At 10–60×, MethPhaser had N50 1.1–1.8-fold that of LongPhase 2 (0.96–5.49 versus 0.87–3.12 Mb), at phased fractions within 0.2 percentage points. LongHap had N50 0.47–2.1-fold that of LongPhase 2 (0.41–5.40 Mb), with shorter blocks only at 10×. MethPhaser and LongHap lengthen blocks by joining them with methylation evidence. Because one wrong join places a whole block on the opposite haplotype, Hamming distance measures this error better than the switch error rate, which counts it as a single switch (Supplementary Fig. 1e). In MethPhaser blocks, 4.0–7.2% of SNVs were on the wrong haplotype, compared with 1.7–5.3% for LongPhase 2 and 1.8–6.9% in the input of MethPhaser before its joins (Supplementary Fig. 3); in LongHap blocks, 4.4–20% were. The longer blocks of MethPhaser and LongHap thus contained more SNVs on the wrong haplotype.

With SNVs, indels, SVs and 5mC, LongPhase 2 had 2.5–3.2-fold lower SNV switch error

rates than WhatsHap in SNV-only mode and 3.0–3.8-fold lower rates than WhatsHap with indels (Fig. 3f). Its blocks were 6–52% longer than those of WhatsHap in SNV-only mode and 6–21% shorter than those of WhatsHap with indels at 10–40× (within 1.4% at 50–60×; Supplementary Table 6). LongHap with the same four variant classes had 2.4–4.5-fold higher switch error rates than LongPhase 2 at 10–60× (Fig. 3f and Supplementary Table 4). At 10–60×, its block N50 was 0.46–1.8-fold that of LongPhase 2 (0.47–6.83 versus 1.02–4.14 Mb), shorter only at 10×, with Hamming distances of 6.4–21.9%, compared with 3.6–7.1% for LongPhase 2.

Each variant class contributed differently to four-class phasing (Fig. 3f, Supplementary Fig. 4e–h and Supplementary Table 6). Adding 5mC to SNVs increased block N50 by 2–16% at 10–60× and kept SNV switch error rates within 3% of those of SNV-only phasing. Adding indels increased N50 by 16–33% but also increased switch error rates by 14–24%. Adding SVs changed N50 by at most 1.7%, because an SV does not join two blocks on its own. With all four classes, blocks were 18–46% longer than with SNVs only (4.14 versus 2.87 Mb at 60×), with 14–27% higher SNV switch error rates. Adding SVs and 5mC to indels changed indel-pair switch error rates by at most 7% (Supplementary Table 2). In four-class phasing, LongPhase 2 phased 97.4–99.2% of the heterozygous 5mC sites that it called, 66.1–68.5% of the heterozygous SVs called by Sniffles2 and 30.5–45.4% of benchmark heterozygous indels (replicate 1; Supplementary Data 1).

Benchmark choice changes the measured separation between phasers

Against GIAB v4.2.1, differences in switch error rates among LongPhase 2, WhatsHap and HapCUT2 narrowed as coverage increased (Fig. 4a). LongPhase 2 made 1,641–1,839 switch errors across coverages, whereas WhatsHap and HapCUT2 decreased from 2,773 and 3,050 at 10× to 1,801 and 2,088 at 60×. The ratio between the highest and lowest switch error rates fell from 1.6 to 1.2–1.3 at 30–60×. When assessed against v5.0q, switch-error counts decreased with increasing coverage for all three tools, including from 1,503 to 477 for LongPhase 2, and the rate ratio increased from 3.3 to 4.3–5.0 (Fig. 4b). Relative to v4.2.1, v5.0q reduced errors for LongPhase 2 but increased them for HapCUT2 at every coverage and for WhatsHap below 60×. Hamming distance changed little (Supplementary Table 7).

Coincident switch errors accounted for much of the convergence under v4.2.1. The fraction of LongPhase 2 switch errors shared by both other tools increased from 63% at 10× to 93% at 60×. At 60×, these errors accounted for 86% of WhatsHap errors and 74% of HapCUT2 errors; corresponding shares under v5.0q were only 38%, 10% and 8% (Fig. 4c). In shared benchmark regions at 30–60×, each tool made 1,133–1,492 switch errors against v4.2.1 but 32–393 against v5.0q (Supplementary Table 8). The truth sets thus differ in phase at many shared sites, although possible v4.2.1 phase-transfer errors have not been checked in the reads.

The separation between phasers also depended on genomic context. Within the v5.0q benchmark regions, LongPhase 2 had the lowest switch error rate of the three tools at every coverage. Rates were 2.1-fold lower than for WhatsHap and 2.6-fold lower than for HapCUT2 at 10×, increasing to 3.3–3.8-fold and 7.8–9.5-fold lower at 30–60× (Supplementary Table 8). Among GIAB genomic strata[26], the separation was largest in segmental duplications (3.2–5.4-fold at 10 and 60×) and smallest outside difficult regions (1.7–2.2-fold; Fig. 4d,e). The relative advantage

of LongPhase 2 was greater in regions covered by v5.0q but not v4.2.1[33]: WhatsHap and HapCUT2 had 4.2–8.3 and 7.1–11 times its switch error rate, compared with 1.7–2.5 and 2.0–9.4 times in regions covered by both benchmarks (Fig. 4d,e). These comparisons involved 18–41 switch errors for LongPhase 2 in the newly covered regions per coverage.

Outside the benchmark regions of both truth sets, where the truth phase is least certain, switch errors accounted for an increasing share of each tool's total as coverage rose. This share increased from 28% to 69% for LongPhase 2 and from 43–46% to 62–75% for the other tools between 10 and 60×. Rates in these regions were 0.8–1.1% for LongPhase 2 and 2.6–4.9% for the other tools. These sites contributed 50–56% of the genome-wide difference in switch-error counts at 10× and 60–77% at 60×. We therefore treat this component of the comparison as exploratory (Supplementary Table 8).

LongPhase 2 phases slightly fewer benchmark SNVs and places fewer genotype errors on haplotypes than WhatsHap

LongPhase 2 phased slightly fewer benchmark heterozygous SNVs than WhatsHap and HapCUT2 despite identical input calls. We measured completeness as the fraction of benchmark heterozygous SNVs phased. WhatsHap and HapCUT2 phased 0.7–1.0 percentage points more than LongPhase 2 at every nanopore coverage. For all three tools, the phased fraction rose steeply up to 16× and plateaued at 91–92% from 20× (Fig. 5a and Supplementary Table 1). On HiFi data, WhatsHap and HapCUT2 phased 0.4–0.5 percentage points more and HiPhase 0.2–0.3 percentage points more, and all four tools increased from 87% at 10× to 92% at 20–50× (Fig. 5c). On nanopore data with indels, WhatsHap phased 2.5–4.5 percentage points more benchmark heterozygous indels, with a larger difference at higher coverage (Supplementary Table 7). On HiFi data with indels, WhatsHap, HapCUT2 and HiPhase phased 2–3, 6–9 and 5–8 percentage points more benchmark heterozygous indels than LongPhase 2 (Supplementary Table 4).

A larger share of the calls phased by LongPhase 2 matched benchmark heterozygous SNVs. Among heterozygous SNV calls phased on chr1–22, the share matching a benchmark heterozygous SNV increased from 89.6% to 93.3% for LongPhase 2 and from 86.2% to 91.5% for WhatsHap (Fig. 5b). The share was higher for LongPhase 2 in every replicate, although the difference narrowed from 3.4 percentage points at 10× to 1.8 at 50–60×. F1, the harmonic mean of this share and completeness, was also higher in every replicate. The difference decreased from 0.9–1.1 percentage points at 10–18× to 0.4 at 50–60× (83.3–92.3% for LongPhase 2 and 82.3–91.9% for WhatsHap; Supplementary Table 1). On HiFi data, the share was also highest for LongPhase 2 at every coverage, increasing from 92.2% to 95.0%, compared with 90.3% to 94.6% for WhatsHap and 91.1% to 93.9% for HiPhase (Fig. 5d and Supplementary Table 3). The difference from WhatsHap narrowed from 1.8 percentage points at 10–20× to 0.3 at 50×, whereas the difference from HiPhase remained at 1.1–1.5 points.

LongPhase 2 also phased fewer genotype errors than WhatsHap. These errors place an incorrect allele on a haplotype when a homozygous site is called and phased as heterozygous[33]. At positions with a homozygous v5.0q genotype on chr1–22, WhatsHap phased 27,135, 13,706 and 11,815 heterozygous calls at 10, 30 and 60×, compared with 14,154, 9,578 and 8,339 for

LongPhase 2 (1.4–1.9 times as many; Fig. 5e). WhatsHap also phased 1.3–1.4 times as many calls with a different allele in v5.0q or no v5.0q record. The latter category included 190,344 calls for WhatsHap and 147,307 for LongPhase 2 at 60×; benchmark absence alone does not establish a genotype error.

Calls phased by only one tool explained these differences (Supplementary Table 9). WhatsHap phased 124,000–151,000 calls that LongPhase 2 left unphased; 15–20% matched benchmark heterozygous SNVs. Among benchmark-matching calls in blocks with at least two assessed variants, 22.8%, 16.2% and 9.9% were phased incorrectly at 10, 30 and 60×, three to six times its overall rate. Conversely, LongPhase 2 phased 34,000–50,000 calls that WhatsHap left unphased. About 90% were absent from the benchmark, and 92–93% lay in segmental duplications or outside one-to-one assembly alignments. Among the 958–1,489 benchmark-matching SNVs assessed in such blocks, 12.7–14.9% were phased incorrectly. Calls phased by either tool alone also overlapped that tool's switch-error intervals 2.4–9.9 times more often than its phased calls overall.

At higher sequencing coverage, calls left unphased by LongPhase 2 alone included more high-depth sites than those left unphased by WhatsHap alone (Supplementary Fig. 5). At 10×, 62–71% of calls in each set had depth below 20×. At 30 and 60×, this remained true for 51–59% of calls left unphased by WhatsHap alone, compared with 12–17% of those left unphased by LongPhase 2 alone. The latter set had a depth peak near the sequencing coverage and included more high-depth calls. At 60×, 28% had depth of at least 90×, and 15.5% reached the variant caller's depth cap. Only 3% of calls left unphased by WhatsHap alone reached 90×.

The excess depth persisted when measured relative to sequencing coverage. At 30 and 60×, 28% and 17% of calls left unphased by LongPhase 2 alone had at least twice the sequencing coverage, compared with 8% and 2% for WhatsHap alone. Such depth can arise when reads from two paralogous copies map to one locus, a known source of false variant calls[33]. At 60×, 3,669 calls left unphased by LongPhase 2 alone had depth of at least 90× within v5.0q benchmark regions; 92.8% were absent from the benchmark VCF and thus classified as false positives by this benchmark. However, calls left unphased by LongPhase 2 alone overlapped segmental duplications or regions outside one-to-one assembly alignments less often than those left unphased by WhatsHap alone (54–75% versus 92–93%). The source of the excess depth remains unresolved.

At 60×, SNVs withheld at high phasing uncertainty lie mostly outside one-to-one assembly alignments

Most SNVs that LongPhase 2 withheld at high phasing uncertainty were absent by position from the v5.0q VCF. LongPhase 2 withheld 61,389–64,201 such SNVs per genome at 10–60× (one replicate per coverage). Among these SNVs, the share absent by position from the v5.0q VCF increased from 84.0% to 93.6% with coverage, while the heterozygous share decreased from 10.9% (6,684) to 3.5% (2,156); the remainder were homozygous (Fig. 5f). Allele matching reduced the heterozygous counts to 6,409 and 1,999. Benchmark absence alone does not establish a call error because the benchmark depends on assembly alignment and variant representation.

An exploratory analysis of T2T-HG002 assembly alignments placed 11.4% of the 62,230

SNVs unphased at 60× in regions where both haplotypes aligned one-to-one to GRCh38. This compared with 92.6% of a random background sample of phased SNVs (odds ratio for lying outside these regions, 98; Fig. 5g and Supplementary Table 10). On autosomes alone, the shares were 9.1% and 92.9% (odds ratio, 131), with enrichment outside one-to-one alignments on every autosome (Fig. 5h). Sites outside the one-to-one alignment regions clustered in pericentromeric satellites, particularly at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 7,072 unphased SNVs within one-to-one alignments, 3,843 (54%) had no assembly variant at the same start coordinate.

This assembly comparison identifies genomic context but does not independently validate phase or allele identity. Coordinate matching can miss variants represented at another position. The background sample also included withheld sites, which biases the comparison towards no difference. The one-to-one regions came from raw assembly alignments and retain assembly errors removed during benchmark curation. Moreover, v5.0q derives from the same assembly. Thus, the analysis extends the assessment beyond benchmark regions, but cannot establish whether the excluded phase assignments were correct.

LongPhase 2 phases a genome faster than WhatsHap and HapCUT2

On one thread, LongPhase 2 phased SNVs 9.6- to 34.2-fold faster than WhatsHap and HapCUT2 with the least peak memory of the three tools, and with 24 threads it phased a 60× nanopore genome in less than one minute (Supplementary Fig. 6 and Supplementary Table 11).

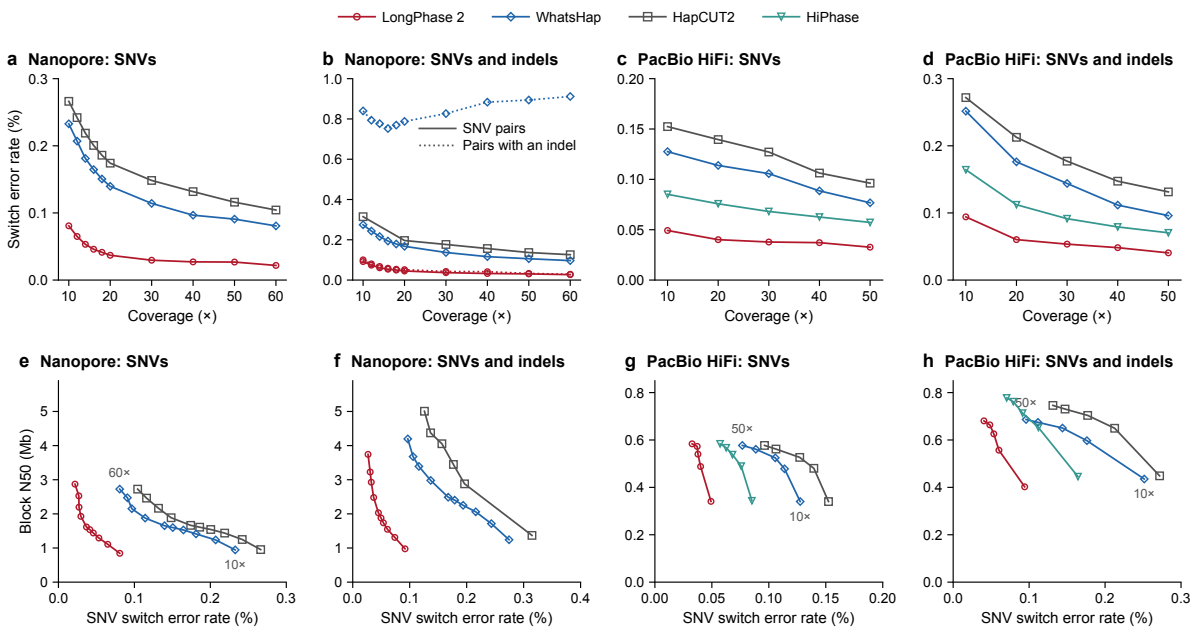


Figure 2: **Accuracy and contiguity of LongPhase 2 compared with other phasers on sequence variants from nanopore and PacBio HiFi data.** HG002 nanopore R10.4.1 data at 10–60× and PacBio HiFi data at 10–50×, scored against the v5.0q benchmark. Top row, SNV switch error rate against coverage; bottom row, SNV switch error rate against phase-block N50 for the same runs, one point per coverage. **a**,**e**, Nanopore SNVs. **b**,**f**, Nanopore SNVs and indels; in **b**, dotted lines show variant pairs that include an indel. **c**,**g**, PacBio HiFi SNVs. **d**,**h**, PacBio HiFi SNVs and indels.

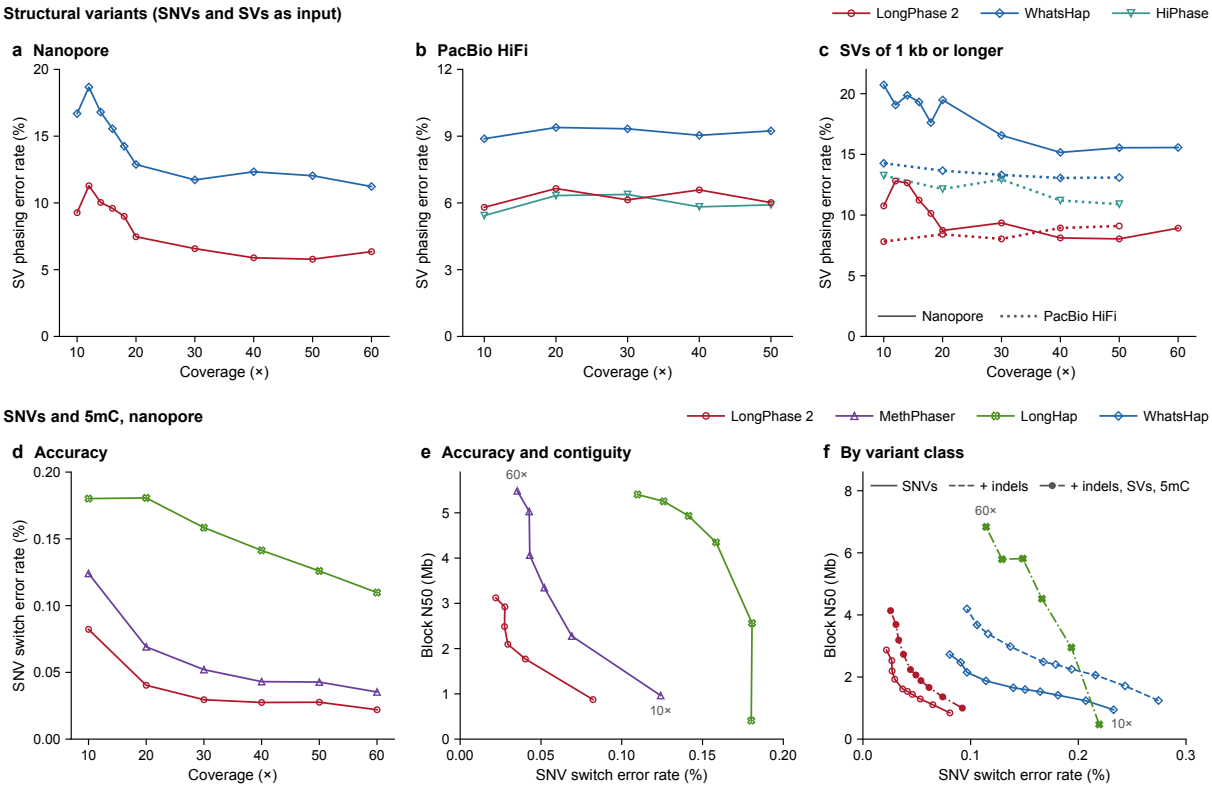


Figure 3: **Joint phasing of SVs and 5mC by LongPhase 2 compared with other phasers.** HG002, replicate 1 unless stated. SV phase was scored against the T2T-HG002 assembly and SNV phase against v5.0q. **a**,**b**, SV phasing error rate against coverage on nanopore (**a**) and PacBio HiFi (**b**) data; SVs shorter than 1 kb are 90–93% of the SVs scored. **c**, SVs of at least 1 kb on nanopore (solid) and PacBio HiFi (dotted) data. **d**, SNV switch error rate against coverage of SNV and 5mC phasing on nanopore data. **e**, SNV switch error rate against phase-block N50 of the runs in **d**, one point per coverage. **f**, The same by variant class; filled markers, SNVs, indels, SVs and 5mC; LongPhase 2 and WhatsHap are means of ten replicates at 10–20×.

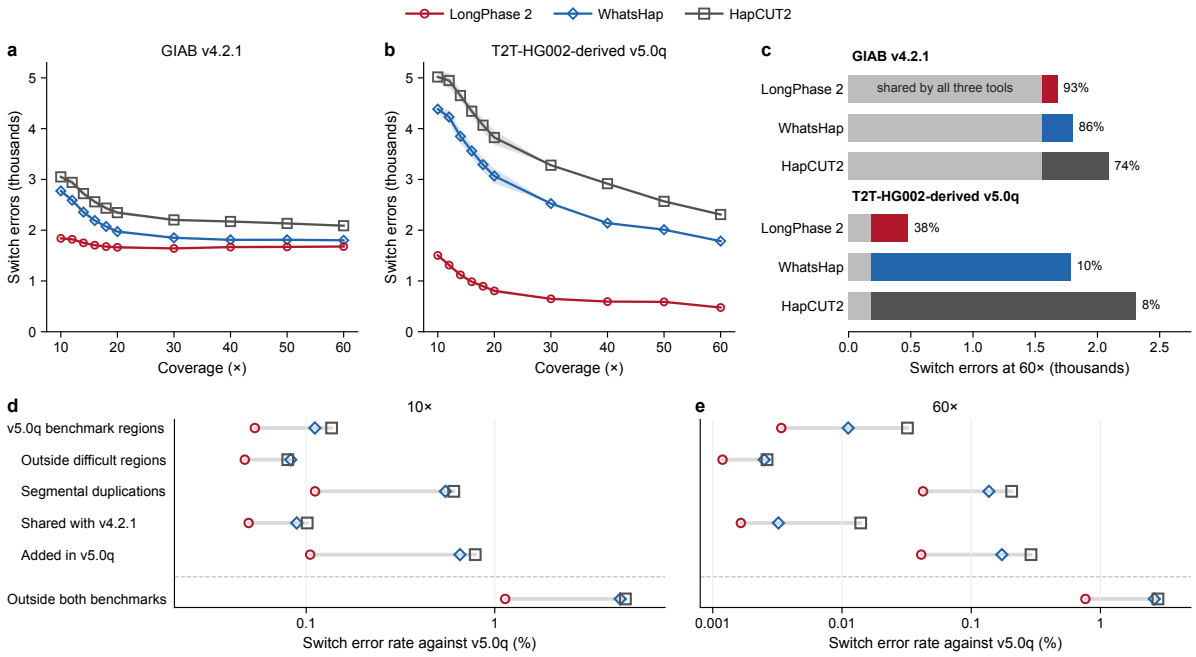


Figure 4: **The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates as in Fig. 2; both truth VCFs used in full. **a**,**b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**). **c**, Switch errors at 60×; grey, switch errors made by all three tools at the same SNV pair, with their share of each tool's total. **d**,**e**, Switch error rate against v5.0q at 10× (**d**) and 60× (**e**) inside the v5.0q benchmark regions and in subsets of them: outside difficult regions and in segmental duplications (GIAB v3.6 strata), and in the parts that v4.2.1 also covers or that v5.0q adds. Below the dashed line, outside the benchmark regions of both truth sets, where the truth phase is least certain (exploratory). A variant pair is assigned to a region when both of its variants lie in it (Supplementary Table 8).

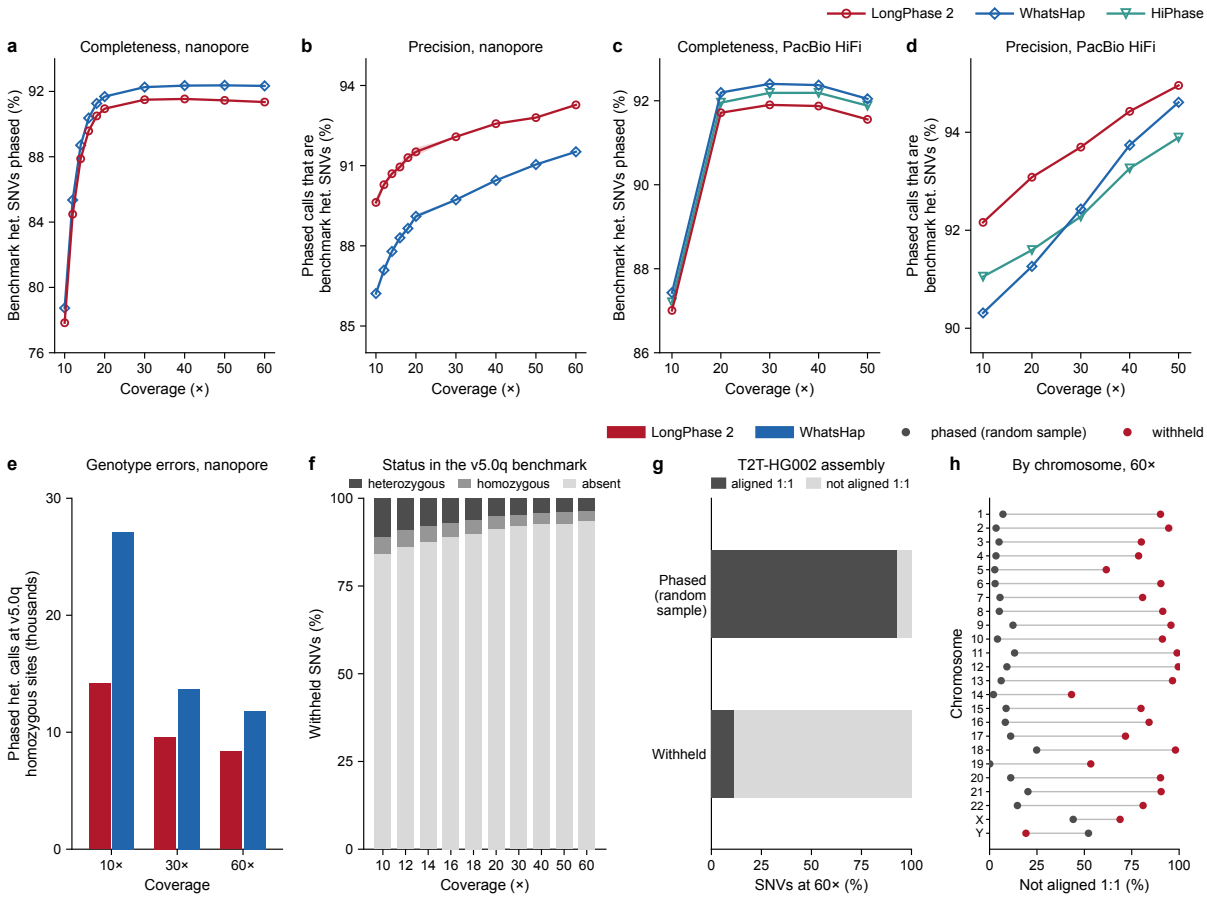


Figure 5: **Calls phased by LongPhase 2, WhatsHap and HiPhase, and SNVs withheld at high phasing uncertainty.** SNV-only phasing of HG002 nanopore R10.4.1 data (**a**, **b**, **e**–**h**) and PacBio HiFi data (**c**, **d**). **a**–**d**, coverages and replicates as in Fig. 2; **e**–**h**, replicate 1. **a**,**c**, Completeness: fraction of the heterozygous SNVs of the v5.0q benchmark phased from nanopore (**a**) and HiFi (**c**) data; every tool received the same heterozygous calls. **b**,**d**, Precision: share of the phased calls on chr1–22 that match a benchmark heterozygous SNV in position and alleles, for nanopore (**b**) and HiFi (**d**) data (Supplementary Tables 1 and 3). **c**,**d** also show HiPhase, which was given the calls restricted to SNVs. **e**, Phased heterozygous calls on chr1–22 at positions where v5.0q has a homozygous genotype (genotype errors; Supplementary Table 9). **f**, SNVs that LongPhase 2 withheld at high phasing uncertainty, by their genotype in the v5.0q benchmark VCF, matched by position without a benchmark BED. **g**, At 60×, these SNVs and a random sample of phased SNVs, inside or outside the regions where both haplotypes of the T2T-HG002 assembly align one-to-one to GRCh38 (exploratory). **h**, The share outside these regions for each chromosome (Supplementary Table 10).

Discussion

LongPhase 2 combines genetic variation and allele-specific methylation in a shared phasing graph with selective withholding of phase assignments predicted to be unreliable. This approach extends methods that use methylation or SVs to connect blocks in regions with few heterozygous SNVs[2,11,12,21,22]. Whereas LongHap adds methylation after sequence phasing[24], LongPhase 2 integrates both during haplotype construction. On the same reads, LongPhase 2 had 1.5–1.8-fold lower switch error rates than MethPhaser, which post-processes an SNV phasing, and 2.2–5.4-fold lower rates than LongHap (Fig. 3d and Supplementary Table 4); for MethPhaser, most of this difference came from GNN correction. SVs placed in the same graph were phased with 1.3–2.3-fold lower error rates than by WhatsHap, and SVs of at least 1 kb with lower error rates than by HiPhase on HiFi data. The variant classes contributed differently: methylation improved contiguity at similar SNV switch error rates, whereas indels produced most of the gains in contiguity and increases in error. The same trade-off in WhatsHap suggests that difficulties in indel representation or modelling may affect both methods. In joint SNV and indel phasing, LongPhase 2 with GNN correction had 8–32-fold lower switch error rates than WhatsHap at variant pairs containing an indel. GNN correction reduced the additional error associated with indels (Supplementary Note 1 and Supplementary Fig. 7). More accurate indel calls could reduce the increase in phasing error associated with indel co-phasing[16,34].

Selective withholding requires a distinction between evidence agreement and confidence in phase. LongPhase 2 reports phasing entropy and weighted vote totals. Unlike model-based confidence estimates from HapCUT2[9] and statistical phasers[5], low entropy indicates agreement among available votes, not necessarily strong evidence or a low error probability. The GNN uses neighbouring graph and sequence features and may detect candidates missed by an entropy threshold alone. However, its added value over simpler withholding rules and HapCUT2's confidence pruning remains untested at matched variant retention. Its onset-prediction score is neither calibrated nor included in the output. These distinctions limit the use of the reported scores as quantitative measures of error risk.

Withholding variants and splitting blocks also change which variant pairs can be assessed. Switch error rates must therefore be considered together with completeness and contiguity. Before GNN correction, LongPhase 2 already had 1.9–2.5-fold lower switch error rates than WhatsHap at similar block N50. Most additional SNV calls phased by WhatsHap did not match a benchmark heterozygous SNV. Among assessable benchmark heterozygous SNVs phased by WhatsHap alone, 10–23% were phased incorrectly, three to six times its overall rate. Benchmark-matching SNVs phased exclusively by LongPhase 2 had a similar error rate, but were 13–26 times fewer. SVs showed the same pattern: given the same SNVs and SVs, WhatsHap phased 0.7–6.3% more SVs correctly than LongPhase 2, but 1.5–2.3 times as many incorrectly. At 60×, most SNVs unphased by the GNN lay outside one-to-one assembly alignments, where our analysis cannot establish whether their phase was correct.

Benchmark choice changed the apparent separation between methods. At 30–60×, the highest and lowest switch error rates differed by 1.2–1.3-fold against GIAB v4.2.1 but by 4.3–5.0-fold against v5.0q. Many apparent errors under v4.2.1 occurred at the same variant pairs across tools, and most disappeared when the same outputs were scored against v5.0q in shared benchmark

regions. All three phasers agreed with v5.0q at most of these sites, including WhatsHap and HapCUT2, which have no parameters trained on either truth set. Errors in v4.2.1 phase transfer are one possible explanation, but shared reads, alignments and variant calls could also contribute to correlated outputs. Independent checks with reads or parental data are needed to distinguish these explanations.

Within v5.0q benchmark regions, LongPhase 2 had 2.1- to 9.5-fold lower switch error rates than WhatsHap and HapCUT2, with the largest separation in newly covered regions. However, much of the genome-wide difference arose outside the benchmark regions of both truth sets, where the phase is least certain. This component remains exploratory. The v5.0q release evaluated variant calls rather than phase[33]. Our results support comparing phasers against assembly-phased truth sets, with phase errors reported separately inside and outside benchmark regions and stratified by genomic context[26]. Reporting which calls receive phase assignments also captures genotype errors propagated onto haplotypes, which phase-error rates alone do not measure.

Generalization remains the main evaluation limit. All data came from HG002. The network was trained on nanopore runs from the same down-sampling series, using labels from the evaluation benchmark. Genome-wide metrics include 17 training chromosomes. On held-out chromosomes, the advantage over WhatsHap and HapCUT2 persisted, but correction removed fewer errors at a higher cost in withheld SNVs. These chromosomes were excluded from GNN training but contributed to earlier tuning of the graph constants. The HiFi results without retraining support transfer between platforms within HG002, but do not establish transfer to other individuals. Evaluation in other individuals against an independent benchmark, with fixed weights and thresholds, is needed. Subsampling variability was not estimated above 20× for nanopore data or at any HiFi coverage.

Further work should test both the scope of co-phasing and its downstream effects. 5mC evaluation measured phased fractions rather than phasing accuracy, and neither SVs nor 5mC sites had supervised training labels. SV phase was scored with one SV caller against variants called from the assembly rather than a curated SV benchmark. Methylation co-phasing was assessed only on autosomes of a male lymphoblastoid cell line[22]; the method assumes diploidy and balanced allele fractions. Compatibility with pipelines that use LongPhase[15,17,19,20] enables tests of selective withholding in haplotype-aware somatic calling. More accurate germline phase could improve tumour-read assignment, but withholding phase may reduce information available to the caller. Varying the germline phaser while fixing the somatic caller would quantify this trade-off. A calibrated per-variant confidence measure could further allow downstream methods to account explicitly for uncertainty in inherited phase.

Methods

Overview of LongPhase 2

LongPhase 2 is a diploid germline phaser that takes long-read alignments, a reference genome and heterozygous variant calls, and returns phased VCFs and haplotagged alignments (Fig. 1). SNVs, small indels, structural variants (SVs) and allele-specific 5-methylcytosine (5mC) observations

are represented together in one weighted graph per chromosome. Initial haplotypes are resolved from weighted multi-neighbour votes, reads are used for a second phase-consistency pass, and a graph neural network (GNN) then scores candidate phase-disagreement onsets. Selected variants are unphased, never flipped; if this disconnects a haplotype block, the surviving components receive updated phase-set identifiers. The detailed module rules are given in Supplementary Methods 1–6. The method extends the graph formulation of LongPhase 1[12].

Input variant calls

Heterozygous SNVs and small indels were taken from PEPPER-Margin-DeepVariant[10]; homozygous and filtered records were carried through unchanged but did not enter the graph. Indels in short tandem repeats were retained as evidence but down-weighted, because a misplaced repeat indel can link the wrong alleles. SVs were taken from Sniffles2[13] or cuteSV[14] with supporting read names; homozygous SVs, duplicated positions and SVs coincident with SNVs were excluded. The exact eligibility and repeat annotation rules are in Supplementary Method 2.

Calling allele-specific methylation

Modification probabilities were read from the `MM`/`ML` tags produced by Dorado[35] and carried through alignment. A CpG was called methylated when its 5mC probability was at least 0.8, unmethylated when it was at most 0.2, and ambiguous otherwise. Forward- and reverse-strand observations were mapped to one CpG coordinate. A site was retained as a candidate allele-specific site when methylated and unmethylated reads were sufficiently balanced and the ambiguous fraction was small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6, \qquad \frac{A_R}{M_R + U_R + A_R} \leq 0.2, \tag{1}$$

where M_R, U_R and A_R are the three read counts. Candidates were then required to co-segregate with a neighbouring heterozygous marker. For a pair of sites, writing RR, RA, AR and AA for the four observed state combinations, linkage required

$$\frac{\max(RR + AA,\ RA + AR)}{RR + AA + RA + AR} \geq 0.9. \tag{2}$$

The same criterion was used for SNV-anchored and methylation-only linking. Only CpG dinucleotides were considered. Weak candidates were admitted in a later expansion pass, and consecutive CpGs were merged into one locus with a shared read list for both strands. The resulting methylated and unmethylated read names and depths were supplied to the phasing graph. The depth bound and expansion rules are in Supplementary Method 1, and the procedure is summarized in Supplementary Fig. 8.

Read-level allele observations and filtering

Alignments with mapping quality below 1, unmapped, secondary or duplicate records were excluded. SNV alleles came from the aligned base and its Phred quality; indel alleles from the CIGAR; SV alleles from a matching CIGAR event, the junction of a split alignment or the caller's

read list; and 5mC alleles from the methylation read lists (Supplementary Fig. 9). Supplementary alignments were retained as evidence, so when two alignments of the same read overlapped by more than 20% of their combined span, the shorter alignment was discarded. Under the nanopore setting, a downstream SNV was removed when two SNVs at most 2 bp apart both lay in homopolymers of length at least three. A copy-number-aware filter identified intervals bounded by excess front- or back-clipping and removed variants whose alternate-supporting reads had an unusually high mismatch load relative to reference-supporting reads. Specifically, a read's mismatch load R_m counted the heterozygous sites in the interval at which it carried the alternate allele, and each variant was scored by

$$\mathrm{MR}(v) = \frac{\bar{R}_m^{\mathrm{ALT}}}{\bar{R}_m^{\mathrm{REF}} + \bar{R}_m^{\mathrm{ALT}}}, \tag{3}$$

where the barred terms are the mean loads of reads supporting the ALT and REF alleles of v. Sites with $\mathrm{MR}(v) \geq 0.7$ were removed from all reads. The three-state interval scanner and its constants are specified in Supplementary Method 2.

Weighted phasing graph

Each heterozygous variant contributed reference and alternate allele nodes. For each read, every allele was connected to the next 35 observed variants, with no edge-distance cap. Variants followed by a gap of more than 300 kb to the next heterozygous variant were skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy. A read contributed weight 1 when both observations had quality at least 12 and weight 0.1 otherwise. Indel and 5mC observations and SV alternate observations were assigned fixed quality 60; reference-allele SV observations were assigned 30. Thus all such observations have full weight at the default threshold, while reference-SV evidence is down-weighted only at a higher threshold.

For a variant pair (u, v), the four allele-pair supports were accumulated as w_{rr}, w_{ra}, w_{ar} and w_{aa}. The two possible pairings have support $P = w_{rr} + w_{aa}$ and $Q = w_{ra} + w_{ar}$, and their ambiguity is $s = \min(P, Q)/\max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order (Supplementary Fig. 10). Each variant cast a vote to each downstream neighbour for the pairing with larger support, except when $s > 0.7$; SNV–5mC pairs used a 0.3 threshold and required at least one read-equivalent of support. Votes had weight 1 by default, 20 for near-unanimous support ($s \leq 0.1$ with total support at least 1, or one pairing below 1 while the other was at least 1), and 0.1 when the voting variant was a tandem-repeat indel. The vote was translated using the already assigned haplotype of the upstream variant, producing totals $h_1(v)$ and $h_2(v)$. When more than three votes had support of at most 1 read-equivalent, and at least one qualifying vote remained, only votes with $s < 0.2$, weight at least 1 and a non-indel voting variant were retained. The larger total assigned the haplotype. A tie beyond the last accepted edge opened a new block; a tie within an existing block left the variant unphased and cast no votes. A variant whose votes all came from SVs, beyond the last variant reached by a vote of a small variant, also opened a new block, so that

an SV was never the only link between two blocks; SVs still voted and were phased within a block. The first variant position defined the phase-set identifier. The constants were fixed by a pre-study sweep on HG002 at 10–60× and were not re-tuned for the reported experiments; the parameter tuning is described in Supplementary Method 3.

Read-based correction and phasing entropy

After the graph assignment, each read was assigned to a haplotype by a weighted majority of its phased SNV and SV alleles (weight 1), indel alleles (weight 0.1) and 5mC alleles (not used for this pass; Supplementary Fig. 11). Reads were retained only with a majority above 0.65 and total evidence above 1. For each variant, the consistency of the two read populations was

$$\rho = \frac{\max(n_1, n_2)}{n_1 + n_2},$$

where n_1 counts haplotype-1 reads carrying REF plus haplotype-2 reads carrying ALT, and n_2 counts the converse. The majority pairing assigned the phase only when $\rho > 0.75$; otherwise the variant was left unphased.

For each phased variant, phasing entropy was calculated from the weighted haplotype totals after the single-read guard:

$$\mathrm{PE}(v) = -\sum_{i \in \{1,2\}} p_i \log_2 p_i, \qquad p_i = \frac{h_i}{h_1 + h_2}. \tag{4}$$

It ranges from 0 for unanimous support to 1 bit for equal support. When $h_1 + h_2 = 0$, PE was set to 0 and the case was identified from the accompanying h_1, h_2 fields; this occurs only at the first variant of a block. PE, h_1 and h_2 were written to the VCF, and accepted voting edges with their accumulated weights formed the GNN input graph.

Graph neural network for phase-error detection

Phasing entropy was used to select candidate windows for refinement. Within each window, the GNN combined phasing evidence, genomic context and graph structure to identify phase-disagreement onsets for selective unphasing, as defined by the training labels. Positive targets include isolated disagreements and the first positions of consecutive disagreement runs. The resulting onset scores are not calibrated probabilities.

The GNN operated on merged SNV/indel, SV and 5mC records (Supplementary Fig. 12). A window of at most 41 variants (82 allele nodes) was opened around every phased variant with PE ≥ 0.8, extending 20 variants on either side (Supplementary Fig. 13).

Each allele node carried 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2, and six transforms of them); genomic context, derived from the reference sequence and input variant positions (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator);

and four one-hot variant-class indicators. Six edge features summarized vote weight, genomic distance, allele matching and phase-set relations. Edges were restricted to the window, symmetrized and supplemented with mean-feature self-loops. Exact definitions and normalizations are in Supplementary Tables 12 and 13.

The network used four GPS graph-transformer layers[36], each combining a local four-head GATv2 branch[37,38] and a global four-head self-attention branch[39], followed by residual normalization and a 128–256–128 feed-forward block. The local GATv2 branch aggregates information along edges of the phasing graph, using edge features to weight the contributions of neighbouring allele nodes. The global self-attention branch allows each node to access information from all nodes within the same window, including those without a direct graph connection. A two-layer classifier produced the two-class softmax output. Architecture and inference details are in Supplementary Method 4, Table 14 and Supplementary Fig. 14.

Only variants no further from the trigger than half the largest trigger distance in the window were scored; the same rule defined the nodes used for training loss. Nodes outside that zone supplied context only. Because the zone is defined on genomic distance and not on variant index, the number of variants it holds depends on the local spacing. The two allele-node scores were averaged, followed by averaging across the windows in which the variant lay inside the scoring zone.

A variant with mean score at least 0.35 was unphased while preserving its genotype. Bridge protection was disabled. If removing these variants disconnected a block, the surviving connected components became separate phase sets (Supplementary Method 4).

Training windows came from the same HG002 nanopore co-phasing runs used for evaluation at 10–20× (six coverages and ten down-sampling replicates, 60 runs). Of 722,116 windows, chr1–14 and chr18–20 supplied training (602,788), chr15–16 validation (67,880), and chr17, chr21 and chr22 testing (51,448); the chromosome sets were disjoint. No test-chromosome window entered gradient updates, batch-normalization statistics or checkpoint selection. Windows contained all four variant classes, but only benchmarked SNV and indel nodes had labels, and SV and 5mC correction relies on the model trained on those labelled small variants. The training and evaluation truth was the v5.0q small-variant benchmark. Training windows were limited to 10–20×, whereas results also included 30–60× and PacBio HiFi data.

Labels described local phase orientation. Within each window and phase set, the first benchmarked variant anchored the orientation; later disagreements were labelled as a single-variant flip or unphase event, or as a break at the first site of a longer run. Benchmark-absent variants and SV/5mC records were masked from the loss but retained as graph context. The released model was trained and applied in binary mode. The binary classifier merged the flip, unphase and break labels into one error class; predicted targets were subsequently unphased. Error labels accounted for 2.96% of labelled central nodes in the training split. Correct nodes were sampled independently in each minibatch at a ratio of at most four per error node, and the resulting cross-entropy used focal weighting[40] with $\gamma = 1.5$ and label smoothing 0.05. Training used haplotype-swap and coverage-scaling augmentations; after a swap, every edge took the vote weight of the allele node it then left. The optimizer, stopping rule and environment are listed in Supplementary Table 15.

We used a normalized net-gain criterion for checkpoint selection, assigning unit benefit to each true-positive detection of a label-defined correction target and unit cost to each false-positive prediction. Checkpoints were selected on chr15–16 by maximizing

$$S = R\left(2 - \frac{1}{P}\right) \tag{5}$$

over 81 score thresholds from 0.01 to 0.99, subject to $R \geq 0.005$, where P and R are precision and recall for the label-defined error class. This expression applies for $P > 0.5$; otherwise the score was $P - 1$. The criterion serves as a proxy for correction utility (Supplementary Method 4). Checkpoint details and classification metrics for validation and held-out chromosomes are reported in Supplementary Table 15.

Two label definitions were trained, each with five random seeds (42, 43, 44, 1337 and 3407): benchmark variants matched to the calls by position, and matched by position, REF and ALT, which leaves calls whose alleles differ from the benchmark unlabelled. All other settings were identical. Each of the ten models was applied to the 60 HG002 nanopore SNV-only runs at 10–20×, and its deployment threshold B was set on the validation chromosomes to the same cost of correction, 6.35 phased SNVs withheld per removed switch error, that of the model of the earlier release v2.1 at $B = 0.30$. At that cost, the position-matched labels removed 23.0% of switch errors on the held-out chromosomes (mean of five seeds; range 22.4–23.9%) and the REF/ALT-matched labels 21.6% (20.8–22.9%), although the latter had higher held-out classification scores (Supplementary Table 16). The position-matched labels were therefore kept, and the released model is the one that removed the most switch errors on the validation chromosomes (seed 42). The threshold selected on the validation chromosomes was 0.34 and was rounded to $B = 0.35$ for deployment. In the released implementation, it removed 34.3% of switch errors at 7.98 withheld SNVs per removed error over the whole genome, and 23.5% at 9.26 on the held-out chromosomes (Supplementary Method 4, Supplementary Note 1, Supplementary Fig. 15e and Supplementary Table 17).

Inference was implemented in C++ without external runtime dependencies; the export procedure and numerical agreement checks are described in Supplementary Method 4.

Haplotagging

Reads were tagged from the phased SNV, indel, SV and 5mC alleles they carried. Each informative observation contributed one vote, and a read was assigned to the majority haplotype when its fraction was at least 0.60; reads whose informative variants belonged to multiple phase sets or lacked a majority were left untagged. Tagged reads received HP, PS and phasing-quality fields. SV alleles were re-derived from the CIGAR and 5mC evidence was matched by read name. Full evidence and quality definitions are in Supplementary Method 5.

Benchmark data

The benchmark used HG002 (NA24385) R10.4.1 reads from the Oxford Nanopore Technologies (ONT) `giab_lsk114_2022.12` release: two PromethION flow cells (PAG65784 and PAG68757), SQK-LSK114, 400 bases per second and 4-kHz sampling, basecalled with Dorado

`dna_r10.4.1_e8.2_400bps_sup@v4.0.0` and `5mCG_5hmCG@v2`; their read N50 values were 29.2 and 29.4 kb. Reads were aligned to the GRCh38 no-alt analysis set by ONT with minimap2 v2.24 (`-y -ax map-ont`), retaining reads with mean base quality ≥ 10. The 60× data set is a random 91.14% subset of the 70.4× alignment (samtools v1.15, `view -s 0.9114`), and the 10–50× data sets were down-sampled from it with samtools v1.23.1[41] `view -s` *seed.fraction*. Ten replicates were used at 10–20× and one at 30–60×; variants were called separately at every coverage and replicate (Supplementary Table 18). SNVs and indels came from PEPPER-Margin-DeepVariant r0.8, which reports unphased genotypes, SVs from Sniffles2 v2.8.0, and allele-specific 5mC sites from LongPhase 2. The seeds were 1–9 and 867 at 10× (fraction 0.1666), 12–21 at 12× (0.200), 22–31 at 14× (0.233), 32–41 at 16× (0.266), 42–51 at 18× (0.300), 1–10 at 20× (0.3333), and 42 at 30, 40 and 50× (0.5, 0.6667 and 0.8333). PEPPER-Margin-DeepVariant used the `-ont_r10_q20` model for nanopore data and `-hifi` for HiFi data. HG002 PacBio HiFi (Revio) reads were taken from a 48× GIAB alignment (Supplementary Table 19), which was used as 50×; 10–40× data sets were down-sampled from it with samtools v1.23.1 `view -subsample` (seed 1; fractions 0.2083, 0.4167, 0.6250 and 0.8333), and variants were called separately at every coverage.

Accuracy was scored with `longphase compare` on autosomes 1–22 against two HG002 truth sets: GIAB v4.2.1[30], in the version with phase transferred from a hifiasm assembly (the v4.2.1 benchmark VCF itself is unphased), restricted to biallelic heterozygous records; and GIAB v5.0q[33], derived from the diploid T2T-HG002 assembly[31], in which every heterozygous variant is phased. The truth VCFs were not restricted to their benchmark regions, and a call was scored only where the truth VCF holds the same REF and ALT alleles in a heterozygous genotype. Both truth VCFs list heterozygous variants outside their benchmark regions (9.3% of the autosomal heterozygous records of v5.0q and 3.4% of v4.2.1), so the error counts include sites that the benchmark regions do not cover, among them satellite and segmental-duplication sequence that the T2T-HG002 assembly resolves[31]. To locate switch errors and to score indels, every consecutive variant pair in an intersected block was recorded with a re-implementation of `longphase compare`, which reproduced its counts exactly in all 1,024 scored runs. A pair was assigned to a region when both variants lay in it; regions were the intersection and the differences of the two sets of benchmark regions, and the GIAB v3.6 genome stratifications[26]. Indel switch errors were counted over consecutive pairs that included at least one indel, and the indel Hamming distance was computed in the block orientation chosen on all variants.

Comparison with other methods

WhatsHap v2.8[7] and HapCUT2 v1.3.4[9] were run on the same alignments and variant calls; cross-tool SNV comparisons used SNV-only phasing, with indel phasing disabled in WhatsHap. LongPhase 1.0[12] and LongPhase 2 were run with the nanopore setting, and GNN correction used PE trigger 0.8, score threshold 0.35, no bridge protection and block splitting. Co-phasing experiments separately evaluated the seven variant-class combinations listed in Supplementary Table 6, with and without correction. Indel co-phasing was also compared with WhatsHap's default indel phasing. Margin v2.3.1[10] and Ralphi[28] were evaluated at 10–20×, with ten down-sampling replicates per coverage; these tools were not evaluated at 30–60×. Their configurations

and that of MethPhaser[22] are listed in Supplementary Table 19. MethPhaser was given the SNV-only output of LongPhase 2 without GNN correction instead of the WhatsHap or HapCUT2 phasing its authors recommend as input. MethPhaser and the joint SNV and 5mC phasing of LongPhase 2 started from the same SNV phasing. Without GNN correction, differences between them reflect how methylation is used, after phasing to bridge existing blocks (MethPhaser) or during phasing as additional graph evidence (LongPhase 2); with GNN correction, they also include its effect. GCphase[42] was excluded from the genome-wide comparison because it ignored the FILTER field and phased homozygous-reference calls as heterozygous variants when given the full input VCF (about 1.2 million such calls in the 10× replicate-1 output). GCphase was not evaluated with input restricted to heterozygous SNVs. The HiFi call sets were phased SNV-only with LongPhase 2 (`-pb`), with and without correction, with WhatsHap, with HapCUT2 and with HiPhase (see below); the GNN weights were trained on nanopore data. Three further phasers were run on replicate 1, at 10–60× in steps of 10× on nanopore data and at 10–50× on HiFi data, with default parameters or the workflow their documentation describes (Supplementary Table 19). On nanopore data, LongHap[24] phased SNVs with and without 5mC and, from the same reads, SNVs, indels and SVs with 5mC; because LongHap requires sequence-resolved alleles, Sniffles2 deletions and insertions were given to it with their allele sequences, and inversions, duplications and breakends were omitted. TinkerHap[43] was run on both platforms and HiPhase[11] on HiFi data. Both phase indels by default and have no SNV-only mode, so each was run on the full call sets and on the call sets restricted to SNVs, the variants that WhatsHap phases with `-only-snvs`. For the comparison with indels, LongPhase 2, WhatsHap and HapCUT2 were also run with indel phasing on HiFi data, and HapCUT2 on nanopore data. These runs were scored in the same way as all others (Supplementary Table 4). WhatsHap was run as `whatshap phase -ignore-read-groups -reference` with `-only-snvs` for SNV comparisons, and HapCUT2 as `extractHAIRS -ont 1` (`-pacbio 1` for HiFi) followed by `HAPCUT2` with default parameters, including its default likelihood-based pruning, on the PASS heterozygous SNVs. HapCUT2 rejects genotypes other than two alleles from 0, 1 and 2, so the HiFi call sets given to it omitted such records (134–204 uncalled reference candidates per coverage and one multi-allelic indel at 50×); none of them was a heterozygous SNV. LongPhase 1.0 was re-run on every down-sampling replicate. All tools ran on the same workstation.

Evaluation metrics

Metrics (Supplementary Fig. 16) were computed by intersecting heterozygous query and truth records by position and REF/ALT alleles, then partitioning the intersection into blocks phased in both files; singleton blocks were ignored. Within each block, switch errors were the Hamming distance between truth and query switch-encoded haplotype strings, and the switch-error rate divided their number by the assessed consecutive SNV pairs. Isolated switches and two-consecutive-switch flips were reported separately. Hamming distance was the block-wise minimum over the two orientations, divided by the number of compared variants; unlike switch rate, it included indels when present in both files, so it is not comparable between configurations with and without indels, which we therefore compared by SNV switch error rate. Because a longer block can contain a longer segment of wrong orientation, block-wise Hamming distance

tends to increase with block length, so it does not compare tools with different block lengths on equal terms. Lower Hamming distances were also observed when LongPhase 2 produced longer blocks than WhatsHap. In SNV-only phasing at 30–60×, LongPhase 2 had 2–5% longer blocks than WhatsHap and 1.4–1.8-fold lower Hamming distances. In joint SNV and indel phasing, the blocks of WhatsHap were 12–33% longer, but its indel Hamming distance was 1.8–2.7-fold higher, and this difference increased with coverage as the difference in block length decreased. MethPhaser lengthens its input blocks by joining them without changing phase within them; these joins increased the Hamming distance 1.8–2.8-fold over that of its input at 20–60×. Phased fraction was the number of truth heterozygous SNVs or indels that were phased by the query divided by all truth heterozygous variants of that class. Block N50 was the span-weighted median of query phase-set spans of at least two variants, pooled over chromosomes. Switch-error positions were exported for the unphased-SNV analysis. The exact block and denominator definitions are in Supplementary Method 6.

SV phase evaluation

SV phase was scored on chr1–22 against the variants of the T2T-HG002 Q100 v1.1 assembly called on GRCh38 by dipcall[31,44], which assign every variant, including SVs, to a haplotype; records with a missing allele on either haplotype were excluded. SVs were called with Sniffles2 v2.8.0 with the reference genome and supporting read names on both platforms. A phased heterozygous deletion or insertion of at least 50 bp matched a truth SV of the same type with a size similarity of at least 70% and a position within 1 kb or, for deletions, a reciprocal overlap of at least 50%. Each block was oriented by the majority of its SNVs, and a matched SV was correct when its oriented phase agreed with the assembly. The SV phasing error rate was the number of incorrectly phased SVs divided by the number of correctly and incorrectly phased SVs. WhatsHap v2.8 (`-indels`) and HiPhase v1.7.0 read SVs only as sequence-resolved records in the variant VCF; for these tools, deletions were therefore given the reference sequence as REF and insertions the inserted sequence as ALT, and both were merged with the SNV (and indel) calls. LongPhase 2 received the unmodified SV calls. Inversions, duplications, breakends, symbolic insertions and records with an unknown reference base (296–580 per run) could not be converted; they were not given to WhatsHap and HiPhase and were not scored for any tool.

Analysis of unphased SNVs

An SNV was counted as unphased by GNN correction when it was phased in the pre-correction output and unphased afterwards. Such sites were classified against the v5.0q benchmark VCF, matched by chromosome and position and without a benchmark BED, as absent, homozygous or heterozygous (Fig. 5f); the heterozygous count was also reported with matching REF and ALT alleles, the rule used by `compare`. The analysis used the 10–60× outputs of replicate 1 with the final GNN model.

At 60×, SNVs unphased by GNN correction were also classified against the T2T-HG002 assembly projected to GRCh38 as outside the 1:1 assembly BED, inside without an assembly variant at the same start coordinate, or inside with one; alleles were not compared. All chromosomes, including chrX and chrY, were analysed, and autosome-only values were derived from

the per-chromosome counts. A random sample of 50,000 SNVs phased before correction, which included sites that GNN correction later unphased, provided the background; enrichment was summarized by chromosome and 1-Mb bins.

To compare LongPhase 2 and WhatsHap, the heterozygous SNV calls with single-base alleles and genotype 0/1, which were identical in the two output VCFs, were grouped by the tool that phased them: both tools, LongPhase 2 only, WhatsHap only or neither; calls phased by WhatsHap only were further split by whether LongPhase 2 had already left them unphased before GNN correction. Each call was classified against the v5.0q chr1–22 VCF, without a benchmark BED, as heterozygous with the same position and alleles, heterozygous with other alleles, homozygous or absent; calls on chrX and chrY were counted separately. The calls phased by a tool were those phased by both tools plus those phased by that tool only; phased calls at positions with a homozygous v5.0q record were counted as genotype errors (Fig. 5e). For every SNV-only run of both tools, and on HiFi data also of HiPhase, the share of the phased heterozygous SNV calls on chr1–22 that matched a benchmark heterozygous SNV (precision) was computed from the `compare` count of matched calls, and the F1 score as the harmonic mean of precision and the phased fraction. For the matched calls phased by one tool only, the phase assigned by that tool was scored as in `compare`, and a call was counted as wrongly phased when its phase was opposite to the majority orientation of its block relative to the truth. A switch-error interval extends from the first SNV of a switch-error pair to the base before the second; the share of the calls phased by one tool only inside that tool's intervals was compared with the share of all heterozygous SNV calls phased by that tool. Calls were also annotated by read depth, by the v5.0q benchmark regions, by GIAB v3.6 segmental duplications and by the one-to-one dipcall alignment (Supplementary Data 1). [Add the script behind Fig. 5f (`analyze_unphased.py`) to the repository; the scripts for Fig. 5b,d,e are in `notes/unphased`.]

Runtime and memory

Wall-clock time, CPU time and peak resident memory of SNV-only phasing were measured on one workstation (AMD Ryzen 9 9950X3D, 16 cores, 249 GB of memory, Ubuntu 24.04, input and output on an NVMe solid-state drive), one run at a time with the page cache cleared before each run. LongPhase 2 was run with 24 threads and on one thread; WhatsHap and HapCUT2 have no thread option and ran on one thread (Supplementary Table 11).

Statistics and reproducibility

Ten nanopore down-sampling replicates at 10–20× were summarized by their mean and standard deviation, while 30–60× and HiFi used one down-sampling per coverage. Subsamples for each platform were drawn from the same source sequencing data set of HG002. The reported standard deviations therefore describe variation due to read subsampling, conditional on that data set, rather than variation among independent sequencing experiments. Subsampling variability was not estimated at 30–60× or for HiFi. Tools compared at each coverage and replicate used the same sampled reads, so fold differences between tools and the effect of GNN correction at 10–20× were also computed within each replicate (Supplementary Table 1). No statistical tests, sample-size calculation, randomization or blinding were used. GNN inference used fixed model

weights and no sampling.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data analysed in this study are publicly available: the GRCh38 no-alt analysis set from NCBI (`https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/`); the GIAB HG002 v4.2.1 phase-transfer VCF (`https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/SupplementaryFiles/`) and the v5.0q benchmark (`https://ftp-trace.ncbi.nlm.nih.gov/giab/ftp/release/AshkenazimTrio/HG002_NA24385_son/v5.0q/`) from the GIAB FTP site; and the HG002 nanopore R10.4.1 reads and alignments from the ONT open-data archive (`s3://ont-open-data/giab_lsk114_2022.12/`) under a CC BY-NC 4.0 licence. The truth files used are listed in Supplementary Table 19.

Code availability

LongPhase 2 is open source under the GNU General Public License v3.0 at `https://github.com/twolinin/longphase`; the repository includes a demonstration data set and workflow. The reported results correspond to release v2.1.2 (commit `6b8f2dd5050028a9546026a3311a7c52bc5389db`). The GNN training code, the trained model and the scripts used for the analyses and figures are available in the same repository.[Add the training, analysis and figure-generation scripts to the public repository under an OSI-approved licence, with development-machine paths parameterized, before submission.]

References

[1] Eisfeldt, J., Ek, M., Nordenskjöld, M. & Lindstrand, A. Toward clinical long-read genome sequencing for rare diseases. *Nature Genetics* **57**, 1334–1343 (2025).

[2] Akbari, V. *et al.* Parent-of-origin detection and chromosome-scale haplotyping using long-read DNA methylation sequencing and Strand-seq. *Cell Genomics* **3**, 100233 (2023).

[3] Logsdon, G. A., Vollger, M. R. & Eichler, E. E. Long-read human genome sequencing and its applications. *Nature Reviews Genetics* **21**, 597–614 (2020).

[4] Kolmogorov, M. *et al.* Scalable Nanopore sequencing of human genomes provides a comprehensive view of haplotype-resolved variation and methylation. *Nature Methods* **20**, 1483–1492 (2023).

[5] Hofmeister, R. J., Ribeiro, D. M., Rubinacci, S. & Delaneau, O. Accurate rare variant phasing of whole-genome and whole-exome sequencing data in the uk biobank. *Nature Genetics* **55**, 1243–1249 (2023).

[6] Porubsky, D. *et al.* Fully phased human genome assembly without parental data using single-cell strand sequencing and long reads. *Nature Biotechnology* **39**, 302–308 (2021).

[7] Patterson, M. *et al.* WhatsHap: weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology* **22**, 498–509 (2015).

[8] Martin, M. *et al.* WhatsHap: fast and accurate read-based phasing. *bioRxiv* (2016).

[9] Edge, P., Bafna, V. & Bansal, V. HapCUT2: robust and accurate haplotype assembly for diverse sequencing technologies. *Genome Research* **27**, 801–812 (2017).

[10] Shafin, K. *et al.* Haplotype-aware variant calling with PEPPER-Margin-DeepVariant enables high accuracy in nanopore long-reads. *Nature Methods* **18**, 1322–1332 (2021).

[11] Holt, J. M. *et al.* HiPhase: jointly phasing small, structural, and tandem repeat variants from HiFi sequencing. *Bioinformatics* **40** (2024).

[12] Lin, J.-H., Chen, L.-C., Yu, S.-C. & Huang, Y.-T. LongPhase: an ultra-fast chromosome-scale phasing algorithm for small and large variants. *Bioinformatics* **38**, 1816–1822 (2022).

[13] Smolka, M. *et al.* Detection of mosaic and population-level structural variants with Sniffles2. *Nature Biotechnology* **42**, 1571–1580 (2024).

[14] Jiang, T. *et al.* Long-read-based human genomic structural variation detection with cuteSV. *Genome Biology* **21** (2020).

[15] Zheng, Z. *et al.* Symphonizing pileup and full-alignment for deep learning-based long-read variant calling. *Nature Computational Science* **2**, 797–803 (2022).

[16] Zheng, Z. *et al.* Accelerated long-read variant calling with Clair3 for whole-genome sequencing. *Bioinformatics* **42** (2026).

[17] Zheng, Z. *et al.* ClairS: a deep-learning method for long-read tumor–normal pair somatic small variant calling. *Nature Methods* **23**, 1507–1520 (2026).

[18] Chen, L. *et al.* ClairS-TO: a deep-learning method for long-read tumor-only somatic small variant calling. *Nature Communications* **16** (2025).

[19] Oxford Nanopore Technologies. epi2me-labs/wf-human-variation: Nextflow workflow for the analysis of human genetic variation from Oxford Nanopore sequencing data. `https://github.com/epi2me-labs/wf-human-variation` (2025). Version 2.x; phasing performed with LongPhase (`-use_longphase`).

[20] Keskus, A. G. *et al.* Severus detects somatic structural variation and complex rearrangements in cancer genomes using long-read sequencing. *Nature Biotechnology* **44**, 247–257 (2025).

[21] Akbari, V. *et al.* Megabase-scale methylation phasing using nanopore long reads and NanoMethPhase. *Genome Biology* **22** (2021).

[22] Fu, Y. *et al.* MethPhaser: methylation-based long-read haplotype phasing of human genomes. *Nature Communications* **15** (2024).

[23] Chen, H., Nie, F., Luo, F. & Wang, J. HapBridge: a methylation-guided approach for correcting switch errors and bridging phased blocks in long-read phasing. *bioRxiv* (2025).

[24] Pfennig, A. & Akey, J. M. Methylation-aware long-read phasing significantly improves genome-wide haplotype reconstruction. *bioRxiv* (2026).

[25] Olson, N. D. *et al.* Variant calling and benchmarking in an era of complete human genome sequences. *Nature Reviews Genetics* **24**, 464–483 (2023).

[26] Dwarshuis, N. *et al.* The GIAB genomic stratifications resource for human reference genomes. *Nature Communications* **15** (2024).

[27] Xue, H., Rajan, V. & Lin, Y. Graph coloring via neural networks for haplotype assembly and viral quasispecies reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35 (2022).

[28] Battistella, E., Maheshwari, A., Ekim, B., Berger, B. & Popic, V. Graph-based deep reinforcement learning for haplotype assembly with Ralphi. *Genome Research* **35**, 2617–2625 (2025).

[29] Zook, J. M. *et al.* An open resource for accurately benchmarking small variant and reference calls. *Nature Biotechnology* **37**, 561–566 (2019).

[30] Wagner, J. *et al.* Benchmarking challenging small variants with linked and long reads. *Cell Genomics* **2**, 100128 (2022).

[31] Hansen, N. F. *et al.* A complete diploid human genome benchmark for personalized genomics. *Cell* **189**, 4857–4875.e31 (2026).

[32] Poplin, R. *et al.* A universal SNP and small-indel variant caller using deep neural networks. *Nature Biotechnology* **36**, 983–987 (2018).

[33] Olson, N. D. *et al.* The Genome in a Bottle HG002 assembly-based variant benchmark set enables comprehensive benchmarking of small and structural variants. *bioRxiv* (2026).

[34] Kolesnikov, A. *et al.* Local read haplotagging enables accurate long-read small variant calling. *Nature Communications* **15** (2024).

[35] Oxford Nanopore Technologies. Dorado: high-performance, easy-to-use, open source basecaller for Oxford Nanopore reads. `https://github.com/nanoporetech/dorado` (2025).

[36] Rampášek, L. *et al.* Recipe for a general, powerful, scalable graph transformer. In *Advances in Neural Information Processing Systems*, vol. 35 (2022).

[37] Veličković, P. *et al.* Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).

[38] Brody, S., Alon, U. & Yahav, E. How attentive are graph attention networks? In *International Conference on Learning Representations (ICLR)* (2022).

[39] Vaswani, A. *et al.* Attention is all you need. In *Advances in Neural Information Processing Systems*, vol. 30 (2017).

[40] Lin, T.-Y., Goyal, P., Girshick, R., He, K. & Dollár, P. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)* (2017).

[41] Danecek, P. *et al.* Twelve years of SAMtools and BCFtools. *GigaScience* **10** (2021).

[42] Luo, J., Wang, J., Zhai, H. & Wang, J. GCphase: an SNP phasing method using a graph partition and error correction algorithm. *BMC Bioinformatics* **25** (2024).

[43] Hartmann, U., Shaham, E., Nathan, D., Blech, I. & Zeevi, D. TinkerHap—a novel read-based phasing algorithm with integrated multimethod support for enhanced accuracy. *GigaScience* **14**, giaf138 (2025).

[44] Li, H. *et al.* A synthetic-diploid benchmark for accurate variant-calling evaluation. *Nature Methods* **15**, 595–597 (2018).